

Title: 900-00440 Floor Anchoring Specification	
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CONFIDENTIAL AND PROPRIETARY INFORMATION

AMAZON ROBOTICS, LLC

300 RIVERPARK DRIVE, NORTH READING, MA 01864

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1 Scope of Document

1.1 Purpose of Document

This document serves as a guide for the selection and installation of floor anchoring hardware for stations, chargers, perimeter guarding, as well as other Amazon Robotics controlled hardware that does not require World Wide Building Design template construction details or site-specific anchorage solutions. Amazon Robotics controlled hardware that requires alteration to the existing concrete floor structure such as existing slab on grade removal and replacement with thickened structural footings, through-bolting of baseplates or pedestals for slab on metal deck applications, or contain hardware subject to cyclic fatigue loading. The technologies which currently require site-specific evaluations are:

- Transfer Cartesian Cell (TCC) / Transfer Cartesian Induct Cell (TCIS)
- Cardinal
- Robin

Follow guidance in 900-00442 – Amazon Robotics Floor, Mezzanine, and Environment Specification for the TCC/TCIS, Cardinal, and Robin. Refer to relevant station or other hardware drawings for install locations.

1.2 Document Implementation

Any deviations from Chapter 2 of this document must be approved by Amazon Robotics Hardware Engineering prior to installation. For detailed information see fastener's AML attachments in Agile.

1.3 Anchor Types

The following concrete anchor types are permissible for use for their intended application and are listed in Chapter 2 of this document:

- Concrete Screw Anchors:
 - Concrete screw anchors are screwed directly into a pre-drilled hole in the concrete. The flutes of the screw cut into the surrounding concrete and provide resistance to pull out. These anchors mechanically damage the surrounding concrete and cannot be removed and reinstalled.
 - If reinstalling a concrete screw anchor, a larger diameter concrete screw anchor must be used by drilling a larger diameter hole and installing a corresponding larger diameter concrete screw anchor. Contact Amazon Robotics Hardware Engineering for further guidance if needed.
- Drop-In Anchors:
 - These anchors are placed into a pre-drilled hole. The drop-in anchor is secured by means of a setting tool which expands the sleeve to create grip within the surrounding concrete. These anchors are internally threaded and the anchoring screws used with the drop-in anchor are permitted to be removed and reinstalled as long as the drop-in anchor remains engaged in the concrete.
 - Drop-In anchors are available in lipped and smooth finish. The lip of the lipped drop-in anchors ensures a flush installation with the finished surface of the surrounding concrete. The smooth drop-in anchors require more careful installation to ensure the drop-in anchor is not set too deep into the concrete which could cause installation issues for the corresponding screw. Contact Amazon Robotics Hardware Engineering for further guidance if needed.
- Wedge Anchors:
 - Wedge anchors are inserted into pre-drilled holes. The wedge of the anchor expands by tightening a nut which secures the anchor into the concrete providing resistance to pull out. Wedge anchors have externally threaded studs for tightening of the nut.

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- If reinstalling a wedge anchor, the anchor may continue to be reused as long as the anchor remains engaged in the concrete and no pullout is evident. Contact Amazon Robotics Hardware Engineering for further guidance if needed.

1.4 Concrete Floors

Concrete anchors require a minimum concrete thickness to allow proper installation. See 900-00442- Amazon Robotics Floor, Mezzanine, and Environment Requirements for more details on concrete requirements for various types of AR floors.

2 Approved Anchors

The anchors approved by Amazon Robotics are found in this chapter and are broken down into the following four tables.

Table 1:

- Includes the anchor catalog which provides the necessary information for concrete hole preparation, anchoring parameters, and anchor fastening. Anchoring parameters include the minimum anchor spacing distances from anchor to anchor measured center to center, and minimum anchor distances from the edge of the concrete floor. Edge of concrete floor by definition includes day joints and relief cuts also known as saw cuts in the concrete.

Table 2:

- Includes the screw and washer catalog for use with the concrete anchors; primarily the drop-in anchors. The drop-in anchors have limited internal threading to accept a certain length of screw. The screw length shall be determined by considering the fixture thickness to be anchored, the washer thickness, and the length of the internal threading of the drop-in anchor. As fixture thickness varies, different screw lengths will be required for proper anchoring. Contact Amazon Robotics Hardware for further guidance if needed.

Table 3:

- Includes the concrete floor types, Amazon Robotics controlled hardware to be anchored, and the Amazon Robotics part number for the anchors for permissible use for anchor kit creation. This table includes the various concrete floor configurations found in the Amazon network of buildings. See Appendix A for impact testing references for anchor use justification.

Table 4:

- Includes ResinDek permissible anchors.

ResinDek-HD: See manufacturer specification for more information; High strength applications include all designs intended to resist drive impact (including perimeter guarding, drive unit arrestors, pod blockers, pod canopies, etc). Low strength applications include chargers and designs which are on the pedestrian side of the perimeter guarding (including



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station frames, platforms, ladders, flow racks, FSS frames, etc). See TSB00206 for detailed information on ResinDek installation. Sample images shown below:

Anchor Identification					Hole Drilling Requirements		Anchor Clearance Requirements				Anchor Fastening Requirements		
Concrete Anchor/Anchor Component	Amazon Robotics Part Number	Manufacturer Name and Anchor Series	Manufacturer Part Number	Anchor Size and Length	Concrete Bit Diameter	Minimum Hole Depth	Fixture Hole Diameter	Minimum Embedment Depth	Minimum Spacing (Center to Center)	Minimum Distance From Edge of Slab (ie Day Joint, Relief Cut)	Installation Tool	Tool Size	Maximum Installation Torque
Screw Anchor	200-01925	Hilti KH-EZ	418055	3/8" x 1-7/8"	3/8"	1-7/8"	1/2"	1.11"	3"	2.62"	Wrench	9/16"	19 ft-lb 24 Nm
Screw Anchor	200-01743	Hilti KH-EZ	418057	3/8" x 3"	3/8"	2-3/4"	1/2"	1.86"	3"	2.92"	Wrench	9/16"	40 ft-lb 54 Nm
Screw Anchor	200-04049	Simpson Strong-Tie Titen HD	THD37300HMG	3/8" x 3"	3/8"	2-3/4"	1/2"	1.77"	3"	3"	Wrench	9/16"	50 ft-lb 68 Nm
Screw Anchor	200-04047	Fischer FBS II	536860	10mm x 80mm	10mm	75mm	12.8mm	65mm	40mm	40mm	Wrench	15mm	74 ft-lb 100 Nm
Screw Anchor	200-02116	Hilti HUS3	2079913	10mm x 80mm	10mm	85mm	14mm	58.6mm	50mm	50mm	Wrench	15mm	Not Listed
Drop-In	200-02479	Hilti HKD-Lipped	376967	10mm x 40mm	12mm	43mm	12mm	(Internal Thread Depth) 10-18mm	80mm	140mm	Setting Tool	HSD-G or HSD-M	11 ft-lb 15 Nm
Drop-In	200-03161	Hilti HKV-Smooth	HKV M10x40	10mm x 40mm	12mm	43mm	12mm	(Internal Thread Depth) 10-15.5mm	200mm	150mm	Setting Tool	HSD-G or HSD-M	11 ft-lb 15 Nm
Drop-In	200-02381	Hilti HDI-Lipped	HDI-L+ 3/8	3/8"-16 x 1-9/16"	1/2"	1-9/16"	12mm	(Internal Thread Depth) 5/8"	3.125"	3.125"	Setting Tool	HST 3/8 or HSD-MM 3/8	11 ft-lb 15 Nm
Drop-In	200-03158	Hilti HDI-Smooth	HDI+3/8	3/8"-16 x 1-9/16"	1/2"	1-9/16"	12mm	(Internal Thread Depth) 5/8"	3.125"	3.125"	Setting Tool	HST 3/8 or HSD-MM 3/8	11 ft-lb 15 Nm
Drop-In	200-03807	Hilti HKD-Lipped	376965	10mm x 30mm	12mm	33mm	12mm	(Internal Thread Depth) 10-12.7mm	120mm	105mm	Setting Tool	HSD-G or HSD-M	11 ft-lb 15 Nm
Wedge Anchor	200-03751	Hilti HSL-4	2237418	10mm x 95mm	15mm	90mm	17mm	70mm	160mm	100mm	Wrench	17mm	18 ft-lb 25 Nm
Wedge Anchor	200-02334	Hilti HST3	HST M8x75/10	8mm x 75mm	8mm	59mm	9mm	47mm	50mm	50mm	Wrench	13mm	15 ft-lb 20 Nm
Wedge Anchor	200-0333	Fastenal SD1	2140515	1/4" x 1-3/4"	1/4"	1-1/4"	9/16"	1-1/8"	2-1/4"	1-3/4"	Wrench	7/16"	4 ft-lb 5 Nm
Wedge Anchor	200-04050	Axelent X-Guard	109932	10mm x 80mm	10mm	40mm	12mm	40mm	80mm	100mm	Wrench	17mm	22 ft-lb 30 Nm
Coil Screw	200-02117	Hilti HCA	252010	1/2" x 3"	1/2"	2-3/4"	9/16"	2-5/8"	2-5/8"	2.1"	Wrench	3/4"	80 ft-lb 108 Nm

Table 1 – Approved Anchors Catalog

Screws for Drop-In Anchors and Washers					Fastening Requirements		
Component	Amazon Robotics Part Number	Manufacturer	Manufacturer Part Number	Screw Length / Washer Thickness	Installation Tool	Tool Size	Maximum Installation Torque in Drop-In Anchors
Screw	201-HXHM10x20AS8.8Z	Fastenal	38654	20mm	Wrench	17mm	11 ft-lb 15 Nm
Screw	201-HXHM10x25AS8.8Z	Fastenal	38655	25mm	Wrench	17mm	11 ft-lb 15 Nm
Screw	TBD	Hilti	UK770620	20mm	Wrench	15mm	11 ft-lb 15 Nm
Screw	201-HXH.38-16x1.13AS5Z	Fastenal	13106	1-1/8"	Wrench	9/16"	11 ft-lb 15 Nm
Washer	202-PLN.38x1.25ASSTDZ	Fastenal	11101273	1/8"	N/A	N/A	N/A
Washer	202-PLNM10ASSTDZ	Fastenal	1140358	2mm	N/A	N/A	N/A
Washer	202-PLNM10ASCZ	Fastenal	1138405	2.7mm	N/A	N/A	N/A

Table 2 – Approved Screws and Washers Catalog

Japan 40mm Maximum Anchoring Depth																				
Anchor Identification				Amazon Robotics Controlled Hardware																
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	≤ 8-ft Perimeter / Work Cell Guard				8-ft ≤ 11-ft Perimeter Guard		Pod Tip Canopy	Station Ladder	Angle Iron	Station Frames	Chargers	Charger Wireways	Artemis	Robin*	PSC	Amnesty Reduction	CrossDock
				Troax	NPSG	Axellent	Natom	Troax	NPSG											
Drop-In	200-03807	Hilti HKD-Lipped	10mm x 30mm	Use Permitted						Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Contact ARHW	Contact ARHW
United Kingdom & Continental Europe 75mm Solid Concrete over Corrugated Steel Decking																				
Anchor Identification				Amazon Robotics Controlled Hardware																
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	≤ 8-ft Perimeter / Work Cell Guard				8-ft ≤ 11-ft Perimeter Guard		Pod Tip Canopy	Station Ladder	Angle Iron	Station Frames	Chargers	Charger Wireways	Artemis	Robin*	PSC	Amnesty Reduction	CrossDock
				Troax	NPSG	Axellent	Natom	Troax	NPSG											
Screw Anchor	200-01925	Hilti KH-EZ	3/8" x 1-7/8"	Use Permitted						Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	
Drop-In	200-02479	Hilti HKD-Lipped	10mm x 40mm										Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted		
Drop-In	200-03807	Hilti HKD-Lipped	10mm x 30mm										Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted		
Continental Europe 100mm Solid Concrete over Hollow Core																				
Anchor Identification				Amazon Robotics Controlled Hardware																
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	≤ 8-ft Perimeter / Work Cell Guard				8-ft ≤ 11-ft Perimeter Guard		Pod Tip Canopy	Station Ladder	Angle Iron	Station Frames	Chargers	Charger Wireways	Artemis	Robin*	PSC	Amnesty Reduction	CrossDock
				Troax	NPSG	Axellent	Natom	Troax	NPSG											
Screw Anchor	200-01925	Hilti KH-EZ	3/8" x 1-7/8"	Use Permitted						Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	
Screw Anchor	200-04047	Fischer FBS II	10mm x 80mm				Use Permitted													
Chemical Anchor	200-04048	EKOPOXY ECO4800	12mm x 50mm				Use Permitted													
Drop-In	200-02479	Hilti HKD-Lipped	10mm x 40mm							Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted		
Drop-In	200-03807	Hilti HKD-Lipped	10mm x 30mm							Use Permitted										
North America 4.5-in (114mm) Solid Concrete over Corrugated Steel Decking																				
Anchor Identification				Amazon Robotics Controlled Hardware																
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	≤ 8-ft Perimeter / Work Cell Guard				8-ft ≤ 11-ft Perimeter Guard		Pod Tip Canopy	Station Ladder	Angle Iron	Station Frames	Chargers	Charger Wireways	Artemis	Robin*	PSC	Amnesty Reduction	CrossDock
				Troax	NPSG	Axellent	Natom	Troax	NPSG											
Screw Anchor	200-01925	Hilti KH-EZ	3/8" x 1-7/8"	Use Permitted						Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	
Screw Anchor	200-01743	Hilti KH-EZ	3/8" x 3"	Use Permitted				Use Permitted		Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	
Screw Anchor	200-04049	Simpson Strong-Tie Titen HD	3/8" x 3"		Use Permitted				Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	
Drop-In	200-02479	Hilti HKD-Lipped	10mm x 40mm							Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted		
Drop-In	200-03161	Hilti HKV-Smooth	10mm x 40mm							Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted		
Drop-In	200-02381	Hilti HDI-Lipped	3/8"-16 x 1-9/16"							Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted		
Drop-In	200-03158	Hilti HDI-Smooth	3/8"-16 x 1-9/16"							Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted	Use Permitted		
Wedge Anchor	200-02334	Hilti HST3	8mm x 75mm											Use Permitted	Use Permitted					
Wedge Anchor	200-0333	Fastenal SD1	1/4" x 1-3/4"											Use Permitted	Use Permitted					
Coil Screw	200-02117	Hilti HCA	1/2" x 3"									Use Permitted								
Wedge Anchor	200-04050	Axellent X-Guard Anchor	10mm x 80mm			Use Permitted														
Ground Level Slab on Grade Specific Application 5.5-in (140mm)																				
Anchor Identification				Amazon Robotics Controlled Hardware																
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	≤ 8-ft Perimeter / Work Cell Guard				8-ft ≤ 11-ft Perimeter Guard		Pod Tip Canopy	Station Ladder	Angle Iron	Station Frames	Chargers	Charger Wireways	Artemis	Robin*	PSC	Amnesty Reduction	CrossDock
				Troax	NPSG	Axellent	Natom	Troax	NPSG											
Wedge Anchor	200-03751	Hilti HSL-4	10mm x 95mm																	Use Permitted

Robin* - DOES NOT INCLUDE ROBIN PEDESTAL ANCHORS WHICH MAY REQUIRE THROUGH-BOLTING FROM FINISHED FLOOR TO FLOOR BELOW. ROBIN IN THIS TABLE INCLUDES THE WORKCELL LIGHT CURTAIN STRUCTURE, AND CONVEYOR STRUCTURE ONLY

Table 3 – Applicable Anchors Depending on Concrete Thickness

Floor Type	Anchor Type	Part Number	Angle Iron	Pallet Backwall	Amnesty Reduction Fixture	Chargers	High Cycle Fatigue ⁸	Hole Requirement (Refer to anchor mfg spec for full details)	Min. Embed Depth	EU/NA
ResinDek – HD	High Strength	200-01885, 201-HXHM10x80AS10.9Z , 200-01888						19mm x THRU B-DECK	N/A	NA / EU
	Low Strength	200-02364, 202-PLNM8ASSTDZ						8mm x THRU B-DECK	N/A	NA / EU
		200-02333, 200-01034						¼" x THRU B-DECK	N/A	NA Only

Table 4 – ResinDek Anchor Selection

3 *Anchor Kits*

Amazon Robotics has various kits within the Agile database with a “690” part number prefix. These kits can be found on station BOM’s with a qty of zero, as a kit with the same suffix as the station part number such as (700-0012-157 Station = 690-0012-157 Anchor kit), or listed in the relationship tab of Agile.

3.1 *Anchor Kit Selection*

There are multiple kits depending on application. The Amazon Robotics Solutions team will pick the appropriate anchor kit for each site based on region, slab thickness, and application and add it to the SBOM in order to drive supply chain demand. Anchor kits are typically organized in sets of 25 or multiples appropriate for the application of a particular station. If a situation exists where there is no kit that contains anchors that fit within this specification, a deviation request shall be submitted to Amazon Robotics Hardware Engineering before installation.

Flat washers should be included for all floor anchors within anchor kits. Extra washers may be included in kits for convenience; however, washers shall not be stacked to make up for a bolt of improper length for safety critical Amazon Robotics controlled hardware which includes perimeter guarding, station ladders, and pod tip canopies. No more than two flat washers shall be stacked for non-safety critical applications. For kits using self-tapping concrete screws, washers must be used for sliding ladders and platforms. Washers may also be used for ARF’s and tote racks but is not a requirement.

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Appendix A

Impact Testing References					
Anchor Identification				Anchor Use and Justification	
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	Troax ≤ 8-ft Perimeter Guard	Pod Tip Canopy and Ladder Rail
Drop-In	200-03807	Hilti HKD-Lipped	10mm x 30mm	Japan Only See: https://tinyurl.com/473wncby	Troax Test TR-21-029A Troax Test TR-21-028A
Screw Anchor	200-01925	Hilti KH-EZ	3/8" x 1-7/8"	Troax Test TR-21-027A Troax Test TR-21-025A	Troax Test TR-21-026A
Anchor Identification				Anchor Use and Justification	
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	Natom ≤ 8-ft Perimeter Guard	
Screw Anchor	200-04047	Fischer FBS II	10mm x 80mm	EU Only See NATOM guarding Impact Test-14-03-2019	
Chemical Anchor	200-04048	EKOPOXY ECO4800	12mm x 50mm	EU Only See NATOM guarding Impact Test-14-03-2019	
Anchor Identification				Anchor Use and Justification	
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	Axelent ≤ 8-ft Perimeter Guard	
Wedge Anchor	200-04050	Axelent X-Guard Anchor	10mm x 80mm	NA Only See ECO-018706	
Anchor Identification				Anchor Use and Justification	
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	NPSG ≤ 8-ft Perimeter Guard	
Screw Anchor	200-04049	Simpson Strong-Tie	3/8" x 3"	NA Only See ECO-018615	
Anchor Identification				Anchor Use and Justification	
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	NPSG ≥ 10-ft Perimeter Guard	
Screw Anchor	200-04049	Simpson Strong-Tie	3/8" x 3"	NA Only See ECO-018846	
Anchor Identification				Anchor Use and Justification	
Concrete Anchor	Amazon Robotics Part Number	Manufacturer Anchor Series	Anchor Size and Length	Troax ≥ 10-ft Perimeter Guard	
Screw Anchor	200-01743	Hilti KH-EZ	3/8" x 3"	NA Only See ECO-019011	

Table 5 – Impact Testing References to Justify Table 3 Content

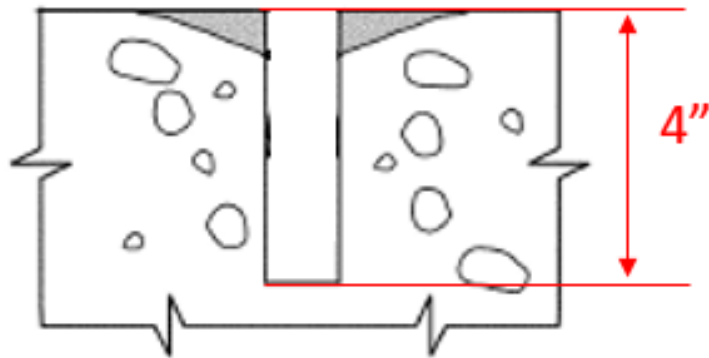
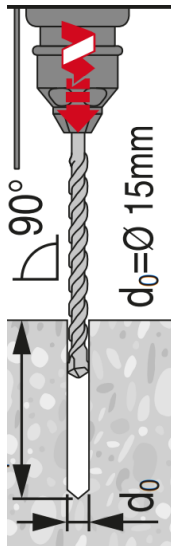
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Appendix B

Installing Heavy Duty Anchors: 200-03751: ANCHOR, WEDGE, HEAVY DUTY, M10. The AML is Hilti with Part Number 2237418. The anchor is shown below.

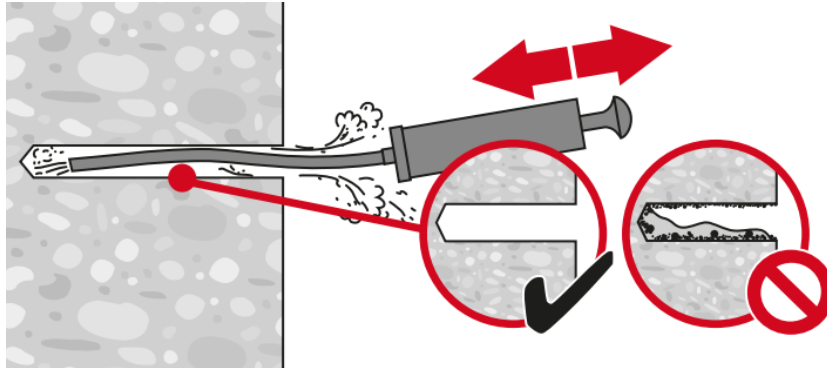


1. Using M15 drill bit, drill 4" minimum into concrete.



2. Using a vacuum, remove any dust that is inside the hole.

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3. Drop anchor into hole

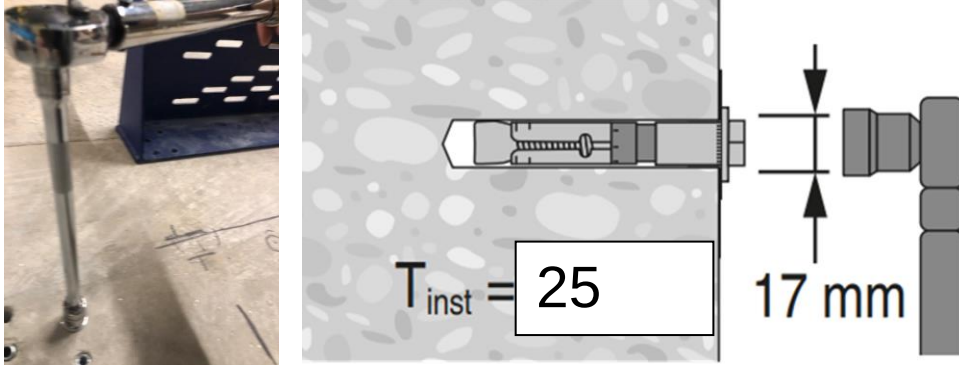


4. Hammer anchor into concrete. It should lay flush with the ground. If it does not, you will need to remove the anchor and drill deeper. **Note: you cannot unscrew the anchor at this point or else components of the anchor will drop into the hole and will be difficult to remove.** Pull the anchor out completely if needed to re-drill.



5. Using a torque wrench fitted with an M17 socket, torque the anchor to 25Nm. **This step is critical to set the anchor.**

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6. Remove the screw and washer from the anchor.



7. Repeat steps 1-6 with remaining holes in part.
8. Place component over anchors. Using a torque wrench, torque the screws to 25 Nm.

Revision History

Revision	Changed Date	Changed By	Description of Change
01	Mar 17, 2016	T. Martin	Initial Release
02	Oct 28, 2016	D. Abbott	Reformatted the overall document; removed 200-01226; added removal and replacement section
03	Dec. 20, 2016	T. Hunter	Updated approved anchors, changed table format
04	Jul. 18, 2017	A. Demicco	Replaced 200-01887 with 201-HXHM10X100AS10.9Z, a longer screw of the same type for ResinDek mounting allowing for use of a steel plate under the decking.
05	Dec. 4, 2017	M. Brady	Updated approved list of anchors to include newly released parts (specifically for chargers). Added information regarding wedge anchors and how they are used.
06	Feb. 13, 2018	J. Jalbert	Updated approved anchor table for ResinDek installs. Changed Pg. 4 note to clarify where to find anchor kit information in PLM system. Added reference image to Pg. 4. Added note to ensure washers are included when creating new anchor kits.
07	Apr. 15, 2020	D. Herzberg	Updated approved anchor table for concrete slab sleeve anchors.
08	July 10, 2020	A. Spaulding	Added 200-03177 (heavy-duty anchors, specifically for CrossDock alignment parts) to list of anchors. Added installation steps for 200-03177. Corrected 200-02116 to be EU Only
09	Nov. 22, 2021	I. Babushkin	Update new anchor part numbers and Hilti (manufacturer) part numbers. Update torque specification related to the new anchors. Changed part number in table 1 (was 200-03177; is 200-03751)
10	Feb. 14, 2022	M. DellaCroce	Reformatted the overall document, incorporated new tables for ease of anchor selection depending on concrete thickness. Incorporated anchor hole requirements, anchor parameters including spacing and distance from edge, as well as fastening requirements. Tables now include an anchor catalog and screw & washer catalog, and impact testing reference table. The Specification now differentiates between Japan, the United Kingdom & Continental Europe, and North America with respect to concrete thickness differences.
11	Jan. 11, 2023	B.D. Willett	Updating anchor spec to include 200-02479 (10mm x 40mm drop in anchor) for 75mm slabs that are not used in cases of impact resistance or life safety. Removing impact resistance and life safety cases from anchor 200-03807.
12	Sep. 9, 2023	M. DellaCroce	Clarified scope of document to ensure delineation between TCC/TCIS, Cardinal, and Robin from AR hardware anchorage recommendations for kitting. Updated table 3 to include work cell guarding and clarified scope of Robin at bottom of table.